

**EDUFYI TECH SOLUTIONS**

Professional Training & Virtual Internships

# MACHINE LEARNING

An abstract geometric pattern consisting of numerous small blue dots connected by thin, light blue lines. The pattern forms a complex, interconnected web of triangles and polygons, resembling a network or a stylized molecular structure. It is positioned in the background, behind the main text and the program type box.

Specialized Industrial Program · 2026 Edition

**PROGRAM TYPE: PROFESSIONAL**

OUR ALUMNI WORK AT

**Microsoft · Capgemini · Deloitte · IBM · Adobe · EY · Bosch**

# WHO ARE WE?

Transforming education through industry-integrated learning

At Edfy Tech Solutions, we are dedicated to transforming the educational journey for college students by offering cutting-edge virtual internships and comprehensive training programs. Our approach combines industry mentorship with practical experience, providing each student with six real-time capstone projects per domain - designed to equip students with the skills to secure their dream jobs upon graduation.

## WHY CHOOSE US?

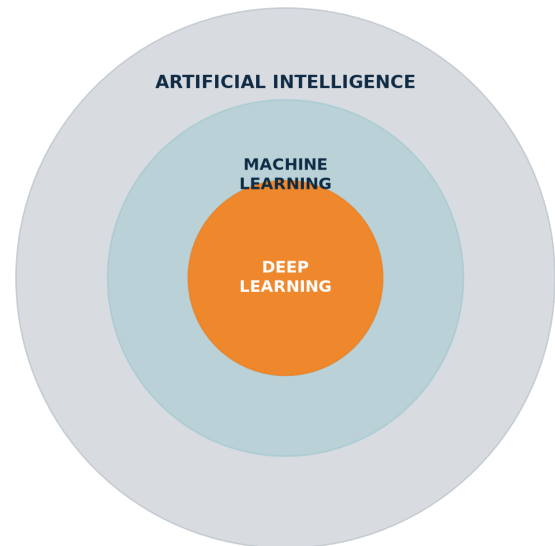
- Choose your speed - master the skills at your pace
- Resume building & aptitude grooming
- 100% placement assistance
- Unlimited mock interviews & GDs
- Wipro DICE ID accredited certifications
- 1:1 counselling sessions
- Certified bootcamps & webinars
- Recorded lectures - anytime, anywhere
- Live mentorship sessions by industry experts
- Industry project playground with real datasets
- Learn from MNC mentors & startup founders
- Exclusive R&D content from leading MNCs
- 24 x 7 dedicated support
- Internships with 150+ partner companies
- Additional certifications: Microsoft, Cisco, IBM, Adobe
- Lifetime personalized LMS access
- Performance-based stipend: skill up & cash in
- Direct access to hiring partners

# WHAT IS MACHINE LEARNING?

The engine of modern data science

Machine Learning is a core component of the growing field of data science. Through statistical methods, algorithms are trained to make classifications or predictions, and to uncover key insights in data mining projects. These insights drive decision-making within applications and businesses, directly impacting key growth metrics.

As big data continues to expand, the market demand for ML engineers and data scientists keeps accelerating - making ML one of the highest-leverage skills of the decade.



*AI > Machine Learning > Deep Learning*

## IN A NUTSHELL

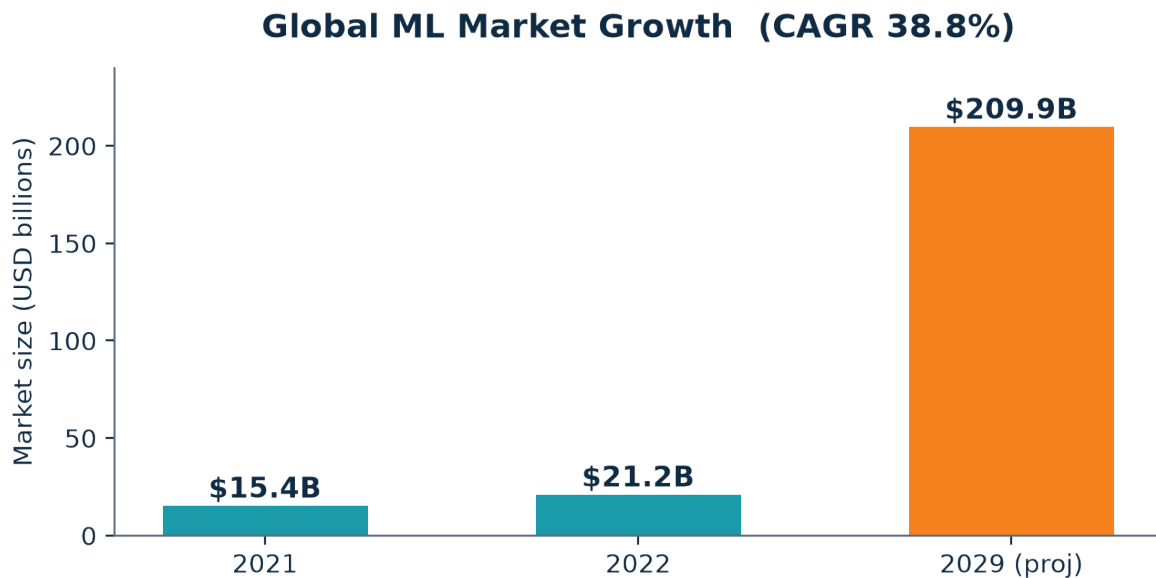
ML systems learn patterns from data instead of following hand-written rules - improving automatically with experience, from spam filters to self-driving cars.

**Data -> Model -> Prediction -> Feedback**

# SCOPE OF MACHINE LEARNING

One of the best career choices of the 21st century

Machine Learning offers abundant high-paying job opportunities, and its future scope is driving a drastic change in the world of automation. The global ML market was USD 15.44 billion in 2021 and is expected to grow from USD 21.17 billion in 2022 to USD 209.91 billion by 2029, at a CAGR of 38.8% during the forecast period.



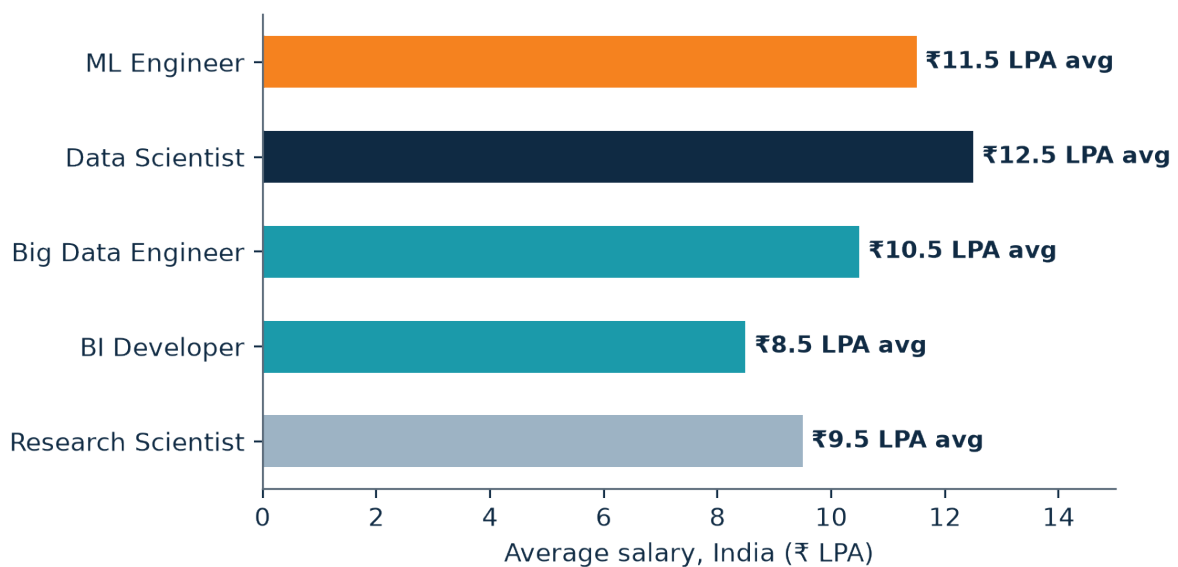
Source: Fortune Business Insights, ML market forecast

# CAREER OPPORTUNITIES

Roles, demand and compensation

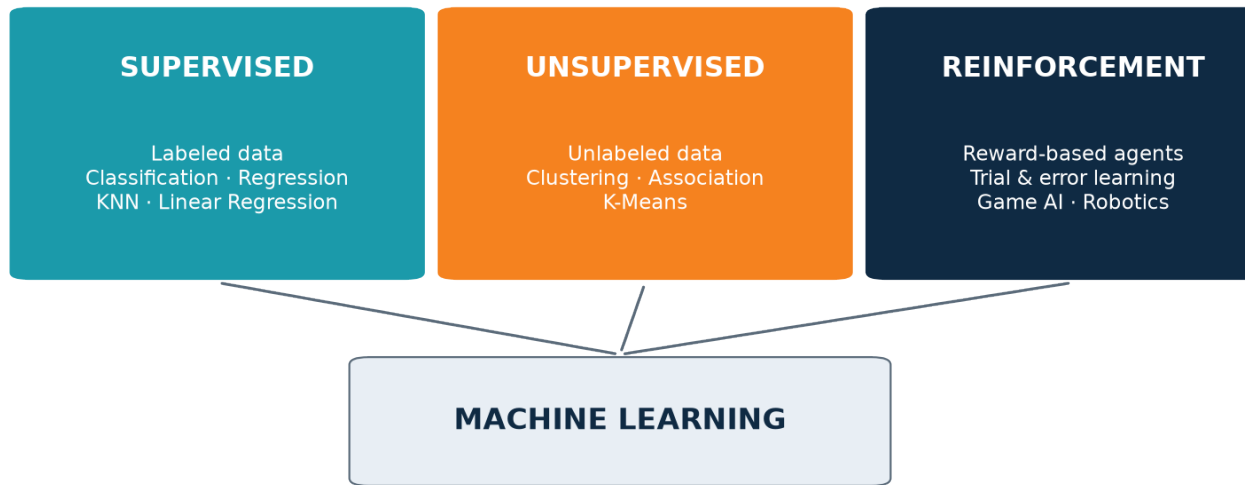
- Machine Learning Engineer
- Data Scientist
- Business Intelligence Developer
- Research Scientist
- Big Data Engineer
- Big Data Architect
- NLP Engineer
- Computer Vision Engineer
- MLOps Engineer

## Career Demand & Compensation Snapshot

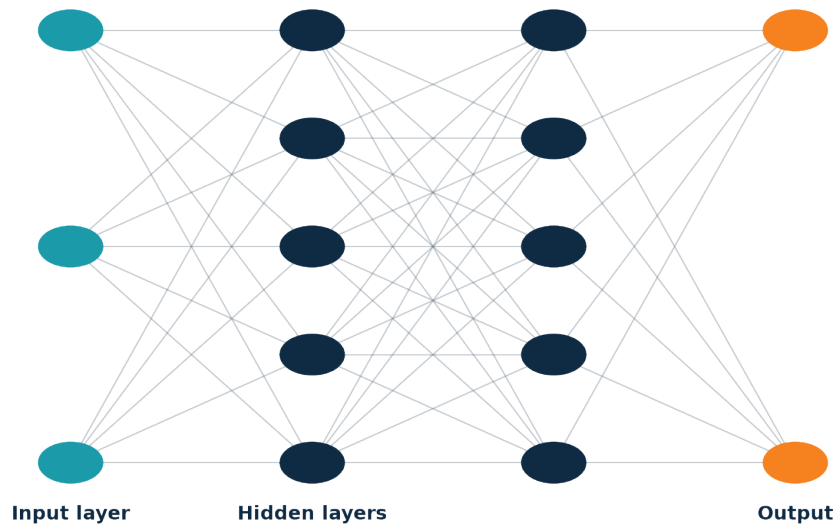


# HOW MACHINE LEARNING WORKS

Paradigms and the neural network model



## INSIDE A NEURAL NETWORK



Weighted connections between perceptrons adjust during training to minimize error.

# PROGRAM CURRICULUM — FOUNDATIONS

Industrial training · Sessions 1–11

## SESSION 1

### Python Kickstart

- Introduction to Python & applications
- Competitive landscape analysis

## SESSION 2

### Python Deep Dive

- Features of Python
- Competitive analysis

## SESSION 3

### Environment Setup

- Download & install Python
- Install Visual Studio Code

## SESSION 4

### Variables in Python

- Python variables
- Writing Python in VS Code
- Using variables

## SESSION 5

### Intro to ML

- What is Machine Learning
- How the human brain takes decisions
- Case: recognizing the letter F

## SESSION 6

### AI vs ML vs DL

- AI vs ML vs Deep Learning
- ML vs AI differences
- DL vs ML vs AI differences

## SESSION 7

### Case Study: Self-Driving Car

- Applications of ML
- Input (data) -> Output (decisions)
- Cases: moving object, static object, curved path

## SESSION 8

### Types of ML

- Supervised ML
- Unsupervised ML
- Reinforcement ML

## SESSION 9

### Supervised ML

- Brain decision-making process
- Supervised learning intro

## SESSION 10

### Classification & KNN

- Classification-based supervised ML
- K-Nearest Neighbours algorithm

## SESSION 11

### KNN in Practice

- How the KNN algorithm works
- Distance calculation walkthrough

# PROGRAM CURRICULUM — CORE ML

Industrial training · Sessions 12–22

## SESSION 12 KNN Logic & Applications

- Logic inside the algorithm
- Example: distance calculation
- Applications of KNN

## SESSION 14 Linear Regression

- What is linear regression
- Regression for supervised ML
- Case: users-per-year data

## SESSION 16 Perceptron Functioning

- Functioning of neural networks
- Threshold value & bias
- Case: school decision-making

## SESSION 18 Bias & Variance

- Limitations of ML
- Bias vs variance comparison

## SESSION 20 Unsupervised ML

- Intro to unsupervised learning
- Supervised vs unsupervised
- Usage of unsupervised learning

## SESSION 22 Reinforcement Learning

- How RL works
- Case study: Flappy Bird game

## SESSION 13

## KNN in Excel

- KNN algorithm in Excel
- Case: distance calculation

## SESSION 15

## Neural Networks Intro

- Introduction to neural networks
- The perceptron & its functions
- Weights in the perceptron algorithm

## SESSION 17

## Training Neural Networks

- Sigmoid activation function
- Training a perceptron
- Error value of the function

## SESSION 19

## Fitting in ML

- Underfitting
- Good fit
- Overfitting

## SESSION 21

## K-Means Clustering

- How K-Means works
- Where K-Means is used

# ADVANCED TRACK — NEW FOR 2026

Beyond the fundamentals: the modern ML stack

## DEEP LEARNING

CNNs for image recognition, RNNs & LSTMs for sequences, transfer learning with pretrained models, regularization & dropout.

## NATURAL LANGUAGE PROCESSING

Tokenization & embeddings, sentiment analysis, transformers & attention, working with Large Language Models (LLMs).

## GENERATIVE AI

Prompt engineering, retrieval-augmented generation (RAG), fine-tuning open models, diffusion models for images.

## COMPUTER VISION

Object detection (YOLO), image segmentation, OCR pipelines, face recognition ethics & applications.

## MLOPS & DEPLOYMENT

Model versioning, Docker & APIs, CI/CD for ML, monitoring drift, deploying on cloud (AWS / Azure).

## RESPONSIBLE AI & XAI

Bias & fairness audits, explainability (SHAP / LIME), data privacy, AI governance and regulation.

**Each advanced module includes a hands-on lab and a portfolio-ready mini project.**

# PROJECTS & ASSESSMENT

Six advanced industry projects · continuous evaluation

## ADVANCE INTERNSHIP PROJECTS

**P1****House Price Prediction**

Regression · feature engineering

**P2****Customer Churn Classifier**

Classification · KNN / logistic models

**P3****Market Basket Segmentation**

K-Means clustering on retail data

**P4****Game-Playing Agent**

Reinforcement learning case build

**P5****Image Classifier (Advanced)**

CNN with transfer learning

**P6****Chatbot with RAG (Advanced)**

LLM-powered document assistant

## QUIZ & EVALUATION

- Module-end quizzes after every session block
- Unlimited mock interviews & group discussions
- Aptitude grooming and resume reviews
- Capstone defense evaluated by industry mentors

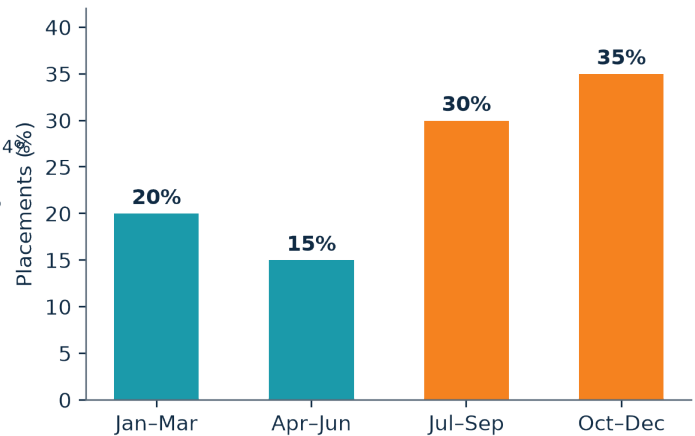
# OUR LEARNER COMMUNITY

Backgrounds, placements and offered profiles



■ UG Management 72.4%  
 ■ UG BTech/BE 15.6%  
 ■ UG Arts/SocSci 4.2%  
 ■ Other 8.0%

## Quarterly Placement Analysis



■ Information Technology 71.6%  
 ■ Business & Mgmt 16.2%  
 ■ Operations 8.1%  
 ■ Core Field 4.1%

### 71.6% of offered profiles are in Information Technology

followed by Business & Management (16.2%), Operations (8.1%) and Core fields (4.1%).

# ACCREDITED CERTIFICATIONS

Industry-recognised credentials for every learner

Every learner receives industry-recognised, accredited certificates on completion of the internship, industrial training and capstone project modules.



**INTERNSHIP CERTIFICATE**  
Accredited Certification



**INDUSTRIAL TRAINING**  
Accredited Certification



**PROJECT COMPLETION**  
Accredited Certification



## INTERNSHIP

Verified internship completion credential



## TRAINING

Industrial training hours certified



## PROJECT

Capstone project completion proof

## WIPRO DICE ID ACCREDITED

Every certificate carries a unique verification ID, verifiable online by recruiters.  
Issued on completion of internship, industrial training and the capstone project.

# GLOBAL MNC CERTIFICATIONS

Certified by the world's leading technology companies

Learners are prepared for and certified through globally recognised professional certification programs from leading technology companies.



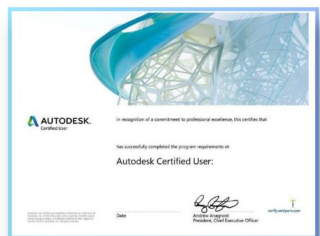
**CISCO**  
Cybersecurity CCST



**MICROSOFT**  
Azure Fundamentals



**IBM**  
Internet of Things



**AUTODESK**  
Certified User



**ESB**  
Entrepreneurship

## EXAM VOUCHERS & PREP INCLUDED

Guided preparation, practice tests and on-campus proctored exams for every certification track.

# AUTHORISED PARTNERS

An authorised certification & testing partner

Edufyi Tech Solutions is an authorised certification and testing partner for the world's leading technology certification bodies.



ADOBE



MICROSOFT



QUICKBOOKS



AUTODESK



CISCO



ESB

## AUTHORISED TESTING & CERTIFICATION CENTRE

Certify on campus with globally accepted credentials.

# CERTIFY. BUILD. GET HIRED.

Certifications you earn · and how to enrol



**Accredited  
Certification**



**Microsoft Azure  
Fundamentals**



**Cisco Cybersecurity  
CCST**



**IBM Internet of  
Things**



**Adobe Certified  
Professional**



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User**

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Enrol in the specialised Industrial Training & Internship program with 150+ partnered companies. Learn from industry experts, ship 6+ real-time projects and earn globally recognised certifications.

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